```
  #type Vertex
  source
  #type Vertex
  dest
  #type double
  weight
  #type string
  label

class Graph
  #type List[Vertices]
  vertices
  #type HashTable[Vertex->List[Edge]]
  adjacencyList

  function Graph()
    adjacencyList ← new HashTable()

  function addVertex(v)
    throw-if v in vertices

vertices.insert(v)
           #1
    adjacencyList[v] ←
[]                                                    #2

  function addEdge(v, u, weight=0, label="")
    throw-if not (v in vertices and u in vertices)
    if areAdjacent(v, u)
then                                                 #3
      removeEdge(v,
u)                                                        #4
    adjacencyList[v].insert(new Edge(v, u, weight, label)
)               #5

  function areAdjacent(v, u)
    throw-if not (v in vertices and u in vertices)
    for e in adjacencyList[v]
do                                                 #6
```